

1. Find the cylindrical coordinates (r, θ, z) and the spherical coordinates of the point with cartesian coordinates $\left(\frac{\sqrt{2}}{8}, \frac{\sqrt{6}}{8}, \frac{\sqrt{2}}{4}\right)$. Explain clearly how you found those coordinates.

For the cylindrical coordinates of the point $\left(\frac{\sqrt{2}}{8}, \frac{\sqrt{6}}{8}, \frac{\sqrt{2}}{4}\right)$, we seek r, θ, z such that

$$\begin{cases} r \cos \theta = \frac{\sqrt{2}}{8}, \\ r \sin \theta = \frac{\sqrt{6}}{8}, \\ z = \frac{\sqrt{2}}{4} \end{cases} \Rightarrow r^2 = \frac{2}{64} + \frac{6}{64} = \frac{8}{64} = \frac{1}{8} \text{ and hence } \begin{cases} \cos \theta = \frac{1}{2}, \\ \sin \theta = \frac{\sqrt{3}}{2}, \\ z = \frac{\sqrt{2}}{4} \end{cases}.$$

Therefore, we find that

$$\boxed{r = \frac{1}{2\sqrt{2}}, \quad \theta = \frac{\pi}{3}, \quad z = \frac{\sqrt{2}}{4}}.$$

For the spherical coordinates of the point $\left(\frac{\sqrt{2}}{8}, \frac{\sqrt{6}}{8}, \frac{\sqrt{2}}{4}\right)$, we seek ρ, θ, ϕ such that

$$\begin{cases} \rho \cos \theta \sin \phi = \frac{\sqrt{2}}{8}, \\ \rho \sin \theta \sin \phi = \frac{\sqrt{6}}{8}, \\ \rho \cos \phi = \frac{\sqrt{2}}{4} \end{cases} \Rightarrow \rho^2 = \frac{2}{64} + \frac{6}{64} + \frac{8}{64} = \frac{1}{4} \text{ and hence } (*) \begin{cases} \cos \theta \sin \phi = \frac{\sqrt{2}}{4}, \\ \sin \theta \sin \phi = \frac{\sqrt{6}}{4}, \\ \cos \phi = \frac{\sqrt{2}}{2} \end{cases}$$

So we obtain that

$$\begin{cases} \sin^2 \phi = \frac{1}{2}, \\ \cos \phi = \frac{\sqrt{2}}{2}, \\ \text{with } 0 \leq \phi \leq \pi \end{cases} \Rightarrow \phi = \frac{\pi}{4} \Rightarrow (\text{from } (*)) \begin{cases} \cos \theta = \frac{1}{2}, \\ \sin \theta = \frac{\sqrt{3}}{2}, \end{cases} \Rightarrow \theta = \frac{\pi}{3}.$$

Therefore, we find that

$$\boxed{\rho = \frac{1}{2}, \quad \theta = \frac{\pi}{3}, \quad \phi = \frac{\pi}{4}}.$$

One could also proceed by inspection, but in that way it is difficult to explain how the coordinates in the new system are obtained! Notice also that, having found the coordinates of the point in cylindrical coordinates, one could use the fact the θ in cylindrical coordinates is the same in spherical and then just find ρ and ϕ

2. Describe the given sets in the system of cartesian coordinates:

- (a) $A : \phi = \frac{\pi}{3}$ in spherical coordinates. Here we have that $0 \leq \theta \leq 2\pi$ which means complete rotation around the z -axis and also ρ is unbounded from 0 to ∞ . Knowing that is a point (x, y, z) belongs to the set A , then ϕ is the angle between the positive z -axis and the line joining the point (x, y, z) to the origin $(0, 0, 0)$. Therefore, the set A is an upward circular cone. To find the cartesian equation of that cone we just use the fact that

$$\tan \phi = \frac{\sqrt{x^2 + y^2}}{z} \Rightarrow z = \sqrt{x^2 + y^2} / \tan\left(\frac{\pi}{3}\right) = \sqrt{\frac{1}{3}(x^2 + y^2)}.$$

- (b) $B : \frac{\pi}{2} \leq \theta \leq \pi$ in cylindrical coordinates.

Here, $z \in \mathbb{R}$ and $r \in [0, \infty)$. We find that the set B is the quarter space

$$x \leq 0, \quad y \geq 0, \quad z \in (-\infty, \infty).$$

- (c) $C : \theta = \frac{\pi}{4}$ in cylindrical coordinates.

The set C corresponds to the half-plane $x = y$ with $x, y \geq 0$ since $x = r \cos \theta$ and $y = r \sin \theta$.

- (d) $D = C$.

3. A torus $\mathbb{T}$ is obtained by rotating a circle C in the xz -plane about the z -axis. If C has radius r and center $(R, 0, 0)$ with $R > r$ then the parametrization of the torus is given by:

$$\vec{r}(s, t) = ((R + r \cos s) \cos t, (R + r \cos s) \sin t, r \sin s) \quad 0 \leq s \leq 2\pi \text{ and } 0 \leq t \leq 2\pi.$$

- (i) Show that the torus $\mathbb{T}$ is a regular surface.

$$\begin{aligned} \vec{r}_s(s, t) &= (-r \sin s \cos t, -r \sin s \sin t, r \cos s), & \text{continuous} \\ \vec{r}_t(s, t) &= (-(R + r \cos s) \sin t, (R + r \cos s) \cos t, 0) & \text{continuous.} \end{aligned}$$

Now either we evaluate $\vec{r}_s(s, t) \times \vec{r}_t(s, t)$ and we show that $\vec{r}_s(s, t) \times \vec{r}_t(s, t) \neq (0, 0, 0)$ for every $s, t \in (0, 2\pi) \times (0, 2\pi)$ or we notice that $\vec{r}_s(s, t) \cdot \vec{r}_t(s, t) = 0$ for every (s, t) and hence,

$$\|\vec{r}_s(s, t) \times \vec{r}_t(s, t)\| = \|\vec{r}_s(s, t)\| \|\vec{r}_t(s, t)\|$$

One can easily find that

$$\begin{aligned} \|\vec{r}_s(s, t)\| &= r, & \|\vec{r}_t(s, t)\| &= R + r \cos s \\ \Rightarrow \|\vec{r}_s(s, t)\| \|\vec{r}_t(s, t)\| &= r(R + r \cos s) \neq 0 & \text{because } r > 0 \text{ and } R > r \text{ while } \cos s \in [-1, 1]. \end{aligned}$$

- (iii) Write down the cartesian equation of the tangent plane to the torus $\mathbb{T}$ at the point $(0, R, r)$.

The given point $(0, R, r)$ corresponds to $s = t = \frac{\pi}{2}$. So the normal vector to the tangent plane is

$$\vec{r}_s\left(\frac{\pi}{2}, \frac{\pi}{2}\right) \times \vec{r}_t\left(\frac{\pi}{2}, \frac{\pi}{2}\right) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & -r & 0 \\ -R & 0 & 0 \end{vmatrix} = (0, 0, -rR).$$

Hence, the tangent plane has cartesian equation $z = r$.

4. Convert in cartesian and cylindrical coordinates the equation of the surface $\mathcal{S}$ given in spherical coordinates by $\rho = 3$.

(i) We recall that

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi \quad \text{with} \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi, \quad \rho > 0$$

from which $\rho^2 = x^2 + y^2 + z^2$. Hence,

$$\rho = 3 \quad \Leftrightarrow \quad x^2 + y^2 + z^2 = 9.$$

So, the surface $\mathcal{S}$ is the sphere centered at $(0, 0, 0)$ with radius 3.

- (ii) From the cartesian equation $x^2 + y^2 + z^2 = 9$, we obtain using $x = r \cos \theta$, $y = r \sin \theta$ and $z = z$ from which $x^2 + y^2 = r^2$ that

$$r^2 + z^2 = 9, \quad \theta \in [0, 2\pi].$$

5. Let $\mathcal{S}$ be the surface described by the vector function $\vec{r}: [0, \frac{3\pi}{2}] \times [\frac{\pi}{4}, \frac{2\pi}{3}]$ defined by

$$\vec{r}(s, t) := (2 \sin s \sin t, 3 \cos t, \cos s \sin t) \quad \forall (s, t) \in \left[0, \frac{3\pi}{2}\right] \times \left[\frac{\pi}{4}, \frac{2\pi}{3}\right].$$

- (a) Show that the surface $\mathcal{S}$ is regular. Notice that

$$\vec{r}_s(s, t) = (2 \cos s \sin t, 0, -\sin s \sin t) \quad \text{and} \quad \vec{r}_t(s, t) = (2 \sin s \cos t, -3 \sin t, \cos s \cos t)$$

which are continuous and

$$\begin{aligned} \vec{r}_s(s, t) \times \vec{r}_t(s, t) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 \cos s \sin t & 0 & -\sin s \sin t \\ 2 \sin s \cos t & -3 \sin t & \cos s \cos t \end{vmatrix} \\ &= (-3 \sin s \sin^2 t, -2 \cos^2 s \sin t \cos t - \sin^2 s \sin t \cos t, 6 \cos s \sin^2 t) \\ &= (-3 \sin s \sin^2 t, -\sin(2t), -6 \cos s \sin^2 t). \end{aligned}$$

Now,

$$\begin{aligned} \vec{r}_s(s, t) \times \vec{r}_t(s, t) = (0, 0, 0) &\Leftrightarrow \begin{cases} -3 \sin s \sin^2 t = 0, \\ -\sin(2t) = 0, \\ 6 \cos s \sin^2 t = 0. \end{cases} \\ &\Rightarrow \begin{cases} t = \frac{\pi}{2} \text{ (from the second equation) and hence } \sin t = 1 \\ \sin s = \cos s = 0 \text{ (from first and second equations)} \end{cases} \end{aligned}$$

and this is impossible. Therefore,

$$\vec{r}_s(s, t) \times \vec{r}_t(s, t) \neq (0, 0, 0) \text{ for every } (s, t) \in \left[0, \frac{3\pi}{2}\right] \times \left[\frac{\pi}{4}, \frac{2\pi}{3}\right]$$

and hence, being $\vec{r}_s$ and $\vec{r}_t$ continuous, we get that the surface $\mathcal{S}$ is regular.

- (b) Write down a cartesian equation of the surface $\mathcal{S}$.

We set

$$x = 2 \sin s \sin t, \quad y = 3 \cos t, \quad z = \cos s \sin t.$$

Looking at these equations, one could notice that the trig. identity $\cos^2 + \sin^2 = 1$ is likely to be the most appropriate. The first attempt is to consider evaluating $x^2 + y^2 + z^2$ which gives

$$x^2 + y^2 + z^2 = 4 \sin^2(s) \sin^2(t) + 9 \cos^2(t) + \cos^2(s) \sin^2(t)$$

Looking at the terms in the left hand side in the order: first, third and second term, we rather consider

$$\begin{aligned} \frac{x^2}{4} + \frac{y^2}{9} + z^2 &= \sin^2(s) \sin^2(t) + \cos^2(t) + \cos^2(s) \sin^2(t) \\ &= (\sin^2(s) + \cos^2(s)) \sin^2(t) + \cos^2(t) = \sin^2(t) + \cos^2(t) = 1. \end{aligned}$$

Hence, the surface $\mathcal{S}$ is piece of the ellipsoid

$$\frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1.$$

- (c) Write down the cartesian equation of the tangent plane to the surface $\mathcal{S}$ at the point $(0, 3/2, \sqrt{3}/2)$.

Let us first seek $(s, t) \in \left[0, \frac{3\pi}{2}\right] \times \left[\frac{\pi}{4}, \frac{2\pi}{3}\right]$ such that $\vec{r}(s, t) = (0, 3/2, \sqrt{3}/2)$. That is

$$\begin{cases} 2 \sin s \sin t = 0, \\ 3 \cos t = \frac{3}{2}, \\ \cos s \sin t = \frac{\sqrt{3}}{2} \end{cases} \Rightarrow \begin{cases} t = \frac{\pi}{3} \quad (\text{second equation}) \\ \sin s = 0, \\ \cos s = 1 \end{cases} \Rightarrow \begin{cases} s = 0, \\ t = \frac{\pi}{3} \end{cases}$$

Now, the normal vector of the tangent plane to the surface $\mathcal{S}$ at the point $(0, 3/2, \sqrt{3}/2)$ is

$$\vec{r}_s(0, \pi/3) \times \vec{r}_t(0, \pi/3) = (0, -\sqrt{3}/2, -9/2)$$

and hence the tangent plane has equation

$$0(x - 0) - \frac{\sqrt{3}}{2}\left(y - \frac{3}{2}\right) - \frac{9}{2}\left(z - \frac{\sqrt{3}}{2}\right) = 0 \quad \Leftrightarrow \quad y + 3z\sqrt{3} = 6.$$

6. Given a differentiable function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$. The normal to the tangent plane of any level surface $f(x, y, z) = k$ at the point (x_0, y_0, z_0) is given by $\nabla f(x_0, y_0, z_0)$.

We have a level surface of the functions $f(x, y, z) = x^2 + 2xy - y^2 + z^2$. So, the normal to the tangent plane at the point $(-1, 2, -3)$ is $\nabla f(-1, 2, -3)$.

$\nabla f(x, y, z) = (2x + 2y, 2x - 2y, 2z)$; so, $\nabla f(-1, 2, -3) = (2, -6, -6)$. Hence, the tangent plane at the point $(-1, 2, -3)$ is

$$2(x + 1) - 6(y - 2) - 6(z + 3) = 0 \quad \Leftrightarrow \quad x - 3y - 3z = 2.$$